RESEARCH UPDATE · NOVEMBER 2026

Decoding along a trajectory

From PIG to Re-Cal, and an edit mode for Nemotron-Labs-Diffusion

Pick a trusted slow decoder, learn when parallel commits stay on its path, and repair the draft instead of throwing it away.

Quentin Fruytier

The University of Texas at Austin · PIG with Anton Xue, Sanjay Shakkottai and Sujay Sanghavi

THE PARADIGM

Trajectory tracking turns parallel decoding into a supervised problem

1 · PICK A REFERENCE

A slow decoder you trust

LLaDA: one token per pass, most confident first. Nemotron-Labs-Diffusion: its own AR mode, greedy.

→

2 · LABEL FOR FREE

Replay it once

Every masked slot at every canvas the walk visits gets a label: does the token it would commit now differ from the one the reference ends with?

→

3 · COMMIT ON A BUDGET

Stay on its path

Predict $\hat{p}_i \approx P[\text{slot } i \text{ disagrees}]$ from the one forward pass, and commit slots while $\Sigma\, \hat{p}_i \le \varphi$.

Self-speculation already tracks the AR trajectory, with an **exact** verifier, so it can only keep a prefix. A **learned** verifier scores every slot, so it can keep tokens anywhere in the block.

RE-CAL · LLADA-8B-INSTRUCT

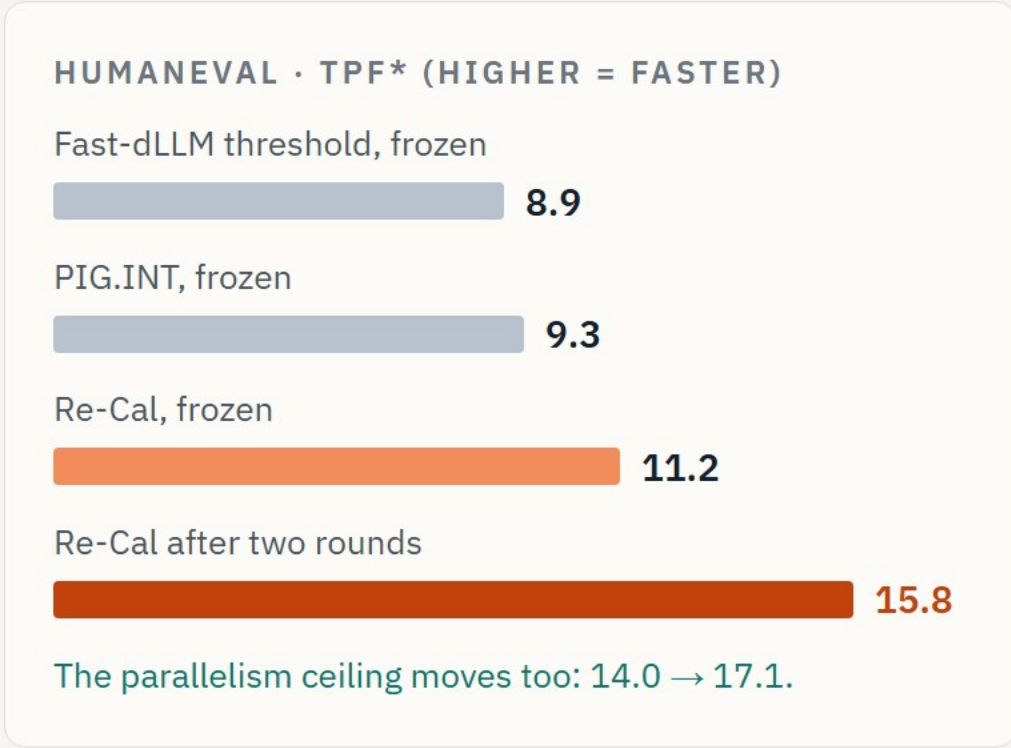
Re-Cal: predict the disagreement directly, then train the model along its own path

$$p_i = g(\text{pass}) \approx P[x_i \neq \text{sequential final}]$$

One small reader on the pass the model already makes. No pairwise entropy matrix, no calibration table: PIG's middle step is gone.

Rule. Commit the most confident slot, then slots in increasing p_i while $\Sigma\, p_i \leq \varphi$.

Rounds. Fine-tune the model along the path Re-Cal itself decodes, refit the reader, repeat (on-path certainty forcing).



164 problems, blocks of 32 · TPF* = fastest setting with pass@1 ≥ 0.322 · the model's own sequential pass@1 stays 0.396

NEW DIRECTION 1 · NEMOTRON-LABS-DIFFUSION-3B SELF-SPECULATION

A PIG-style length adapter prunes quadratic self-speculation to a few candidates

1. **Quadratic self-speculation** verifies the draft and, in the same forward, drafts a fresh block after every possible accept length c = 0…31: j + 1 tokens per forward, at about 1,050 input positions.
2. **A small adapter** on the drafter's hidden states predicts where verification will stop: S(c) = P[accept length ≥ c].
3. **Keep only C blocks**, chosen each step to maximise the expected tokens per forward.

decoder	positions	HumanEval	GSM8K-500
linear SS (release)	2 × 32	3.22	4.12
quadratic SS, all 32	1,047	4.58	5.90
pruned, C = 5	**187**	**4.27**	**5.53**
pruned, C = 3	121	3.98	5.16

Tokens per forward (TPF); every forward counted, recovery passes included

C = 5 is lossless on both sets and gains +32% / +34% TPF: 76–85% of the quadratic gain at 18% of its width.

Strict verification: every committed token is the AR's · paired 95% accuracy intervals vs linear SS contain 0

NEW DIRECTION 2 · EDITING THE DRAFT

Editing is two steps: accept any subset, then re-mask the rest and run again

STEP 1 · ACCEPT ANY SUBSET

Hopper | and | Hopper | are | fast

Keep every slot that agrees with the reference, including the tail **past** the first wrong one.

→

STEP 2 · RE-MASK THE REST, FORWARD AGAIN

Hopper | and | **Blackwell** | are | fast

Only the rejected slot is regenerated, now conditioned on the kept tail on both sides.

Self-speculation

The subset is the verified prefix; the tail is thrown away and redrafted.

Any-subset decoding

The subset may sit past the first rejection; the rest stays masked.

Edit mode

Any subset kept, the rest regenerated in the forward that verifies: the tail survives.

Illustrative example · the Nemotron-Labs-Diffusion paper names non-prefix commits as an open direction

THE PROPOSAL

An edit mode: a fourth mode for Nemotron-Labs-Diffusion

MODE 1

AR

Exact, one token per forward.

MODE 2

Diffusion

Drafts a whole block in parallel.

MODE 3

Self-speculation

Diffusion drafts, AR accepts the prefix.

MODE 4 · NEW

Edit

Reads the drafted block fully visible; per slot, keep or replace.

Bit-identical when off

A per-token-gated LoRA (r = 64, about 99M parameters), never merged: the three shipped modes are unchanged, checked bit for bit.

Lossless with verification

In the same forward, the AR stream commits the exact prefix and the edited tail becomes the next draft.

Labels for free

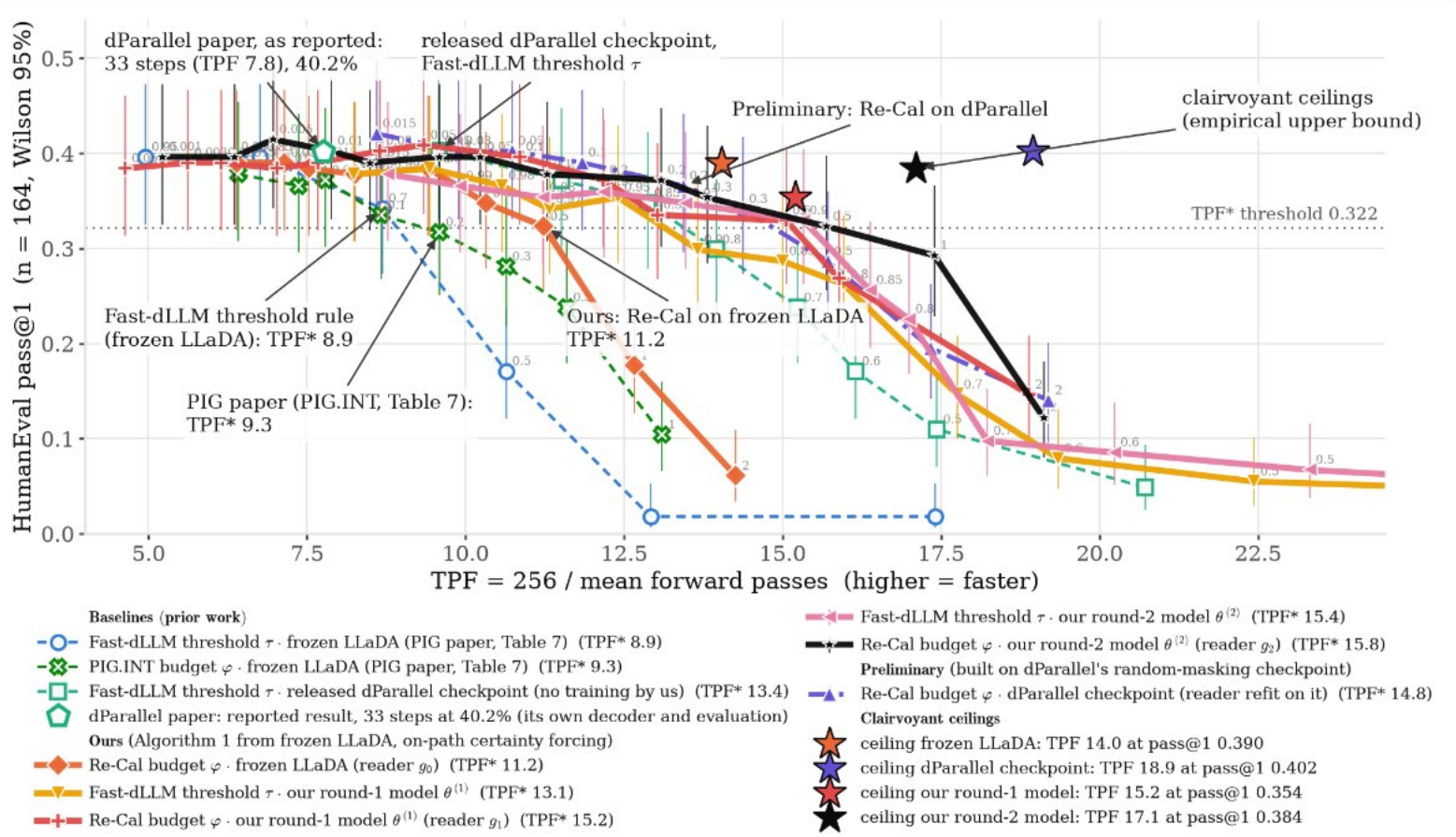
The target is the AR greedy trajectory, read off the lossless loop: prompts only, no ground truth.

Status. Stage 1 is training offline on 2,000 prompts from Nemotron's SFT sets; next, next-draft acceptance on held-out states, then the decode loop.

Nemotron-Labs-Diffusion: Fu et al. 2026, arXiv 2607.05722 (AR, diffusion and self-speculation in one network)

BACKUP · RE-CAL ON LLADA-8B-INSTRUCT

The HumanEval frontier: baselines, Re-Cal and two rounds



Fast-dLLM, frozen

8.9

Re-Cal, frozen

11.2

Re-Cal, round 2

15.8

Stars: clairvoyant ceilings, 14.0 → 15.2 → 17.1.

TPF* · 164 problems, BL = 32, Wilson 95% bars · dParallel lanes are its released checkpoint, decoded in our harness

BACKUP · WHERE THE REMAINING SPEED IS

On a frozen model the bottleneck is the model, not the reader

14.0

Clairvoyant ceiling: commit every slot whose argmax already equals its final sequential token.

0.945

Reader AUC on that path; it commits 77–81% of the committable slots per canvas.

36 vs 34%

A top-10 re-ranker barely beats the runner-up: the rest is not readable from one pass.

dParallel: certainty forcing on random maskings

Most training canvases lie off any path a decoder takes, so the model hedges between outputs, the discrete version of crossing paths.

Ours: on-path certainty forcing

One pairing per prompt, the output Re-Cal reaches at its largest safe budget, and training only along the path it takes: a reflow round.

Frozen LLaDA-8B-Instruct, HumanEval · sequential pass@1 0.396; ceiling pass@1 0.390

BACKUP · THE EXPECTED-VALUE CANDIDATE SET

Each step keeps the candidates with the most expected tokens per forward

$$(N, K) = (j + 1, 2) \text{ if } j < c_1 \text{ miss: plus a draft-only forward}$$

$$(N, K) = (c_i + 1, 1) \text{ if } c_i \le j < c_{i+1} \text{ continue from the block at } c_i$$

$$\mathcal{C}* = \text{argmax over } |\mathcal{C}| = C \text{ of } E[N] / E[K],\ E[K] = 2 - S(c_1)$$

j = accept length, modelled by the adapter's survival S; recomputed every step from that step's draft.

Greedy is enough

The exact optimum (chain dynamic program plus Dinkelbach) is within 0.04 TPF of greedy.

The rule matters

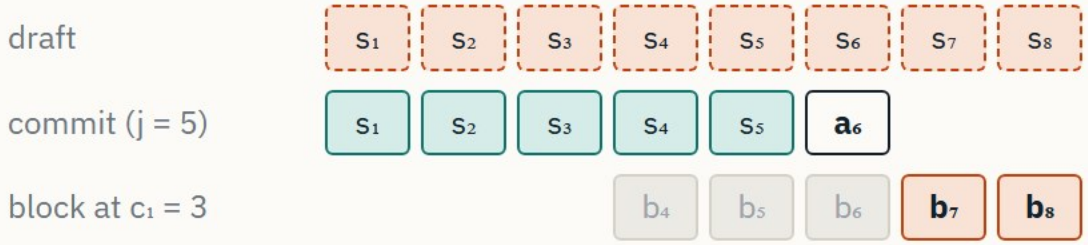
At equal C it beats tuned offsets and survival levels by 4–6% TPF, in all eight cells.

C = 1 is the weak spot

One candidate misses on 26–29% of steps; the recovery forwards cancel the gain.

BACKUP · FIXING C = 1: ANCHORED CANDIDATES

Commit every verified token, and take the next draft from the best anchor



Block 3 was drafted in the same forward from $s_1 \ldots s_3$. Its guesses past a_6 become the next draft, 3 tokens stale instead of 6.

Always commit j + 1. Every AR-verified token is kept whatever the candidates are.

A miss costs no forward. A block at c = 0 is always drafted (Packed-SS); if $j < c_1$ it takes over.

Choose the anchor. $c_1 = \text{argmax}\ c \cdot S(c)$: the expected freshness gained.

Runs in progress.

Packed-SS: Chen and Cai, arXiv 2610.33379 · strict verification, so the output is the AR greedy walk

BACKUP · TRAINING THE EDIT MODE

Trained to predict the AR's token from the visible draft

$$L(\theta) = E[\ -\ \Sigma_\ell\ w_\ell\ \log\ p_\theta\ell(x_\ell\ |\ z,\ c)\ ]$$

$w_\ell = \exp(-d_\ell / \gamma)$, d_ℓ = distance from the commit point

z: the drafted block, fully visible. c: the committed context (cached by the frozen base). x: the AR greedy tokens at the block's positions. Cross-entropy on drafted slots only.

At the optimum, keep-probability $\kappa_\ell = p_\ell(z_\ell) = 1 - P$[slot disagrees | features]: Re-Cal's calibrated reader, plus an edit.

States. A committed prefix of the AR walk, a drafted block after it, the walk's tokens there: from the lossless anchored loop, plus loop-free blocks of 16, 32 or 64 slots.

Data. 2,000 prompts (600 math, 600 chat, 800 code) from Nemotron's SFT sets; 100 held out.

Model. LoRA r = 64 on attention and MLP of all 26 layers; base and LM head frozen; AdamW 1e-4, two epochs.

Borrowed from published DLM editors: own-draft corruption, loss on drafted slots only. **New:** the label-free AR target, the gated fourth mode, repair inside the lossless verify forward.

Recipe from ProSeCo, ME-DLM, CDLM, SCOPE, PRISM, DEdit and others · γ swept over 8, 15, ∞